

Question ID: 2df8f293

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	Medium

Question

Each vertex of a **14**-sided polygon is labeled with one of the **14** letters ***A*** through ***N***, with a different letter at each vertex. If one vertex is selected at random, what is the probability that the letter ***D*** will be at the selected vertex? (Express your answer as a decimal or fraction, not as a percent.)

Correct Answer: .0714, 1/14

Rationale

The correct answer is $\frac{1}{14}$. If one vertex of the polygon is selected at random, the probability that the letter ***D*** will be at the selected vertex is equal to the number of vertices labeled with the letter ***D*** divided by the total number of vertices. It's given that each vertex is labeled with one of the **14** letters ***A*** through ***N***, with a different letter at each vertex. It follows that there is **1** vertex labeled with the letter ***D***. It's also given that the polygon is **14**-sided. It follows that there are a total of **14** vertices. Thus, the probability that the letter ***D*** will be at the selected vertex is $\frac{1}{14}$. Note that 1/14, .0714, and 0.071 are examples of ways to enter a correct answer.